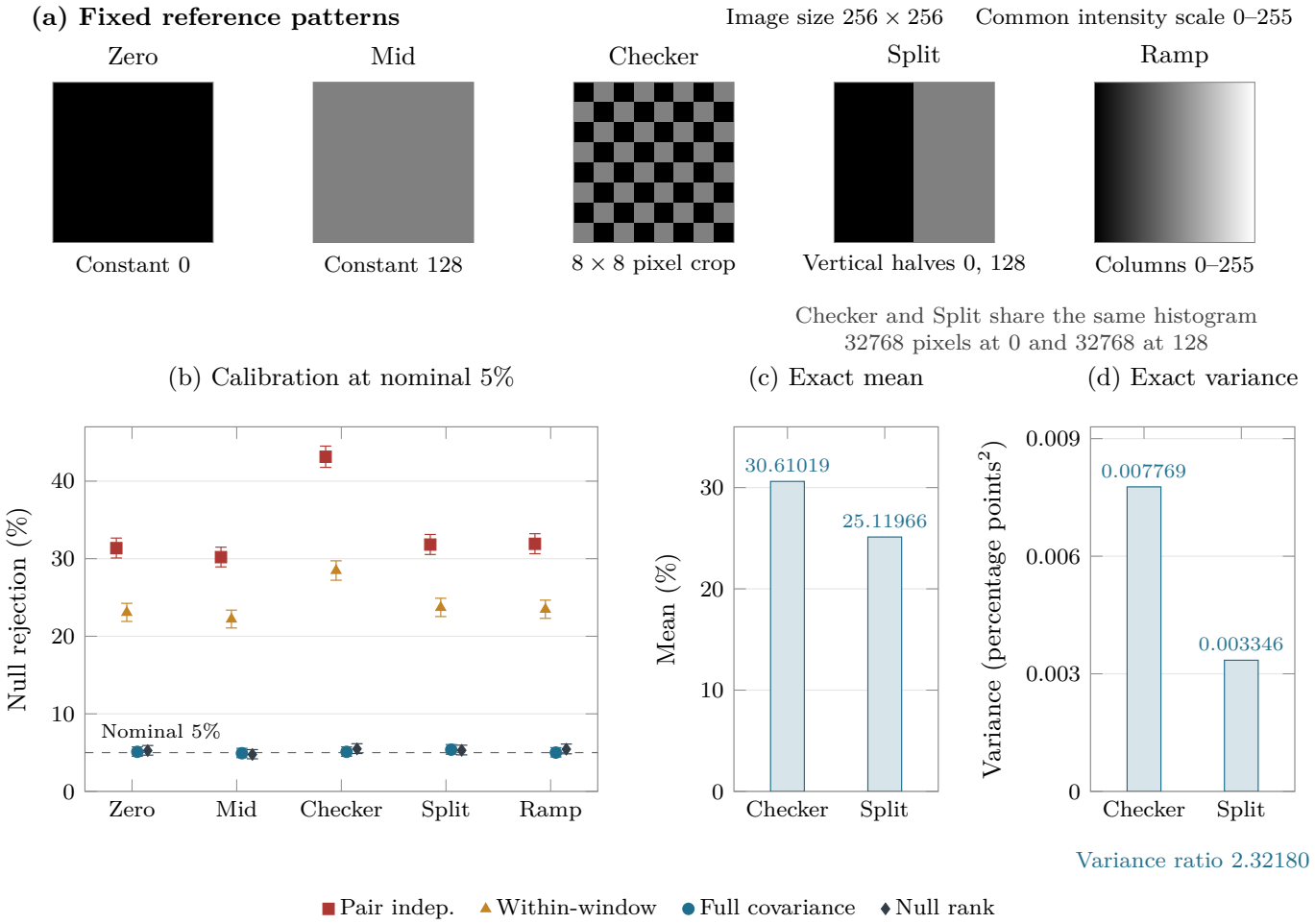


Spatial conditioning in blocked average changing intensity (BACI) with overlapping 2×2 windows, a fixed reference image and $q = 256$. Here B denotes the normalized BACI statistic. (a) Five deterministic 256×256 reference patterns on a common intensity scale. Checker shows an enlarged 8×8 pixel crop; the other thumbnails represent the full images. (b) Rejection at nominal 5% over 5000 independent comparison images per pattern, with mutually independent uniform pixels. Pair indep. denotes the Gaussian rule using the pair-occurrence independence approximation [1], which treats every pair occurrence as independent. Within-window denotes the within-window covariance diagnostic. Full covariance denotes the full-covariance Gaussian rule. Error bars are Wilson 95% intervals, conditional on the realized independent bank of $L = 9999$ reference statistics for the independent-null rank rule. Panels (c,d) display exact $\mathbb{E}[100B]$ and $\mathrm{Var}(100B)$, respectively, for Checker and Split. The mean is expressed as a percentage and the variance in squared percentage points. These patterns have identical histograms, and the checkerboard variance is 2.32180 times the split variance. The moment panels contain deterministic calculations.

Reference

[1] Y. Zhang, Statistical test criteria for sensitivity indexes of image cryptosystems, *Information Sciences* 550 (2021) 313–328. DOI 10.1016/j.ins.2020.10.026.